

APZ lab1 report new

verify that all three required containers are running and maps their respective ports (8080, 8001, 8002)

```
logging-service micro-basics/logging-service "sh -c 'uvicorn serv_..." logging-service About 8 minutes ago Up About 8 minutes 0.0.0.0:8001->8001/tcp, [...]8001->8001/tcp
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
counter-service micro-basics/counter-service "sh -c 'uvicorn serv_..." counter-service 2 minutes ago Up 2 minutes 0.0.0.0:8002->8002/tcp, [...]8002->8002/tcp
facade-service micro-basics/facade-service "sh -c 'uvicorn serv_..." facade-service 2 minutes ago Up 2 minutes 0.0.0.0:8000->8000/tcp, [...]8000->8000/tcp
logging-service micro-basics/logging-service "sh -c 'uvicorn serv_..." logging-service 2 minutes ago Up 2 minutes 0.0.0.0:8001->8001/tcp, [...]8001->8001/tcp
```

POST requests injecting funds and processing deductions:

```
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== POST alice +100 ===" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"alice","amount":100}' ; echo
{"transaction_id":"177783232508558-43baa0e5","balance":100.0}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== POST alice -30 ===" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"alice","amount":-30}' ; echo
{"transaction_id":"1777832329645758-c13577fa","balance":70.0}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== POST bob +250 ===" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"bob","amount":250}' ; echo
{"transaction_id":"177783233313655-844ed53a","balance":250.0}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== POST bob -50 ===" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"bob","amount":-50}' ; echo
{"transaction_id":"1777832343922921-1551f28d","balance":200.0}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== POST carol +500 ===" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"carol","amount":500}' ; echo
{"transaction_id":"1777832347936321-da8135b3","balance":500.0}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== GET /accounts ===" ; curl -s http://localhost:8000/accounts ; echo
{"balances":{"alice":70.0,"bob":200.0,"carol":500.0}}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== GET /user/alice ===" ; curl -s http://localhost:8000/user/alice ; echo
{"user_id":"alice","balance":70.0,"transactions":[{"transaction_id":"177783232508558-43baa0e5","user_id":"alice","amount":100.0}, {"transaction_id":"1777832329645758-c13577fa","user_id":"alice","amount":-30.0}]}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== GET /user/bob ===" ; curl -s http://localhost:8000/user/bob ; echo
{"user_id":"bob","balance":200.0,"transactions":[{"transaction_id":"177783233313655-844ed53a","user_id":"bob","amount":250.0}, {"transaction_id":"1777832343922921-1551f28d","user_id":"bob","amount":-50.0}]}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ echo "=== GET /timings ===" ; curl -s http://localhost:8000/timings ; echo
{"logging_total_seconds":0.832295282888177394,"counter_total_seconds":0.827729481888215148,"logging_calls":7,"counter_calls":0,"logging_avg_ms":4.613611714311056,"counter_avg_ms":3.4661751258268935}
julyfyjulyfy-Micro-ANSIS-57:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$
```

he internal behavior of the individual services by outputting their container logs

```

julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker logs --tail 25 facade-service
INFO: Started server process [7]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.18.0.1:53246 - "POST /transaction HTTP/1.1" 200 OK
INFO: 172.18.0.1:53256 - "POST /transaction HTTP/1.1" 200 OK
INFO: 172.18.0.1:50400 - "POST /transaction HTTP/1.1" 200 OK
INFO: 172.18.0.1:60942 - "POST /transaction HTTP/1.1" 200 OK
INFO: 172.18.0.1:60944 - "POST /transaction HTTP/1.1" 200 OK
INFO: 172.18.0.1:45124 - "GET /accounts HTTP/1.1" 200 OK
INFO: 172.18.0.1:45134 - "GET /user/alice HTTP/1.1" 200 OK
INFO: 172.18.0.1:45148 - "GET /user/bob HTTP/1.1" 200 OK
INFO: 172.18.0.1:37282 - "GET /timings HTTP/1.1" 200 OK
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker logs --tail 25 logging-service
INFO: Started server process [7]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
INFO: 172.18.0.4:48050 - "POST /log HTTP/1.1" 200 OK
INFO: 172.18.0.4:48050 - "POST /log HTTP/1.1" 200 OK
INFO: 172.18.0.4:48050 - "POST /log HTTP/1.1" 200 OK
INFO: 172.18.0.4:55730 - "POST /log HTTP/1.1" 200 OK
INFO: 172.18.0.4:55730 - "POST /log HTTP/1.1" 200 OK
INFO: 172.18.0.4:45898 - "GET /log/user/alice HTTP/1.1" 200 OK
INFO: 172.18.0.4:45910 - "GET /log/user/bob HTTP/1.1" 200 OK
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker logs --tail 25 counter-service
INFO: Started server process [7]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
INFO: 172.18.0.4:53684 - "POST /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:53684 - "POST /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:53684 - "POST /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:37892 - "POST /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:37892 - "POST /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:37892 - "GET /balance HTTP/1.1" 200 OK
INFO: 172.18.0.4:37892 - "GET /balance/alice HTTP/1.1" 200 OK
INFO: 172.18.0.4:44136 - "GET /balance/bob HTTP/1.1" 200 OK
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$

```

restart fully

```

julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker compose restart
[*] Restarting 3/3
  ✓ Container logging-service Started
  ✓ Container facade-service Started
  ✓ Container counter-service Started
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ sleep 4
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ curl -s http://localhost:8000/accounts ; echo # should print ("balances":[])
{"balances":[]}
julfy@julyfy-Nitro-ANS15-57:~/Documents/6th_tern/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$

```

simulate 10 clients to uniquely assigned accounts:

```
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ python3 perf_test.py --scenario A --clients 10 --txns 10000
=== Scenario A: distinct accounts ===
clients=10, txns/client=10000, total=100000
client 0 (runA-user0): 2500/10000
client 8 (runA-user0): 2500/10000
client 7 (runA-user2): 2500/10000
client 9 (runA-user9): 2500/10000
client 1 (runA-user1): 2500/10000
client 2 (runA-user2): 2500/10000
client 6 (runA-user6): 2500/10000
client 4 (runA-user4): 2500/10000
client 3 (runA-user3): 2500/10000
client 5 (runA-user5): 2500/10000
client 0 (runA-user0): 5000/10000
client 8 (runA-user0): 5000/10000
client 7 (runA-user7): 5000/10000
client 9 (runA-user9): 5000/10000
client 6 (runA-user6): 5000/10000
client 1 (runA-user1): 5000/10000
client 2 (runA-user2): 5000/10000
client 4 (runA-user4): 5000/10000
client 5 (runA-user5): 5000/10000
client 3 (runA-user3): 5000/10000
client 7 (runA-user7): 7500/10000
client 8 (runA-user8): 7500/10000
client 0 (runA-user0): 7500/10000
client 6 (runA-user6): 7500/10000
client 9 (runA-user9): 7500/10000
client 2 (runA-user2): 7500/10000
client 1 (runA-user1): 7500/10000
client 4 (runA-user4): 7500/10000
client 5 (runA-user5): 7500/10000
client 3 (runA-user3): 7500/10000
client 7 (runA-user7): 10000/10000
client 0 (runA-user0): 10000/10000
client 8 (runA-user8): 10000/10000
client 9 (runA-user9): 10000/10000
client 6 (runA-user6): 10000/10000
client 4 (runA-user4): 10000/10000
client 1 (runA-user1): 10000/10000
client 2 (runA-user2): 10000/10000
client 5 (runA-user5): 10000/10000
client 3 (runA-user3): 10000/10000
wall time = 272.72 s
successful txns = 100000
failed txns = 0
throughput = 366.7 req/s
logging cumulative = 1658.57 s (avg 16.59 ms over 100000 calls)
counter cumulative = 1598.17 s (avg 15.98 ms over 100000 calls)
final balances = {'runA-user0': 10000.0, 'runA-user0': 10000.0, 'runA-user7': 10000.0, 'runA-user1': 10000.0, 'runA-user2': 10000.0, 'runA-user3': 10000.0, 'runA-user5': 10000.0, 'runA-user4': 10000.0, 'runA-user9': 10000.0, 'runA-u
ser9': 10000.0}
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$
```

```
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ curl -s http://localhost:8000/tinings | python3 -m json.tool
{
  "logging_total_seconds": 1658.5675303408002,
  "counter_total_seconds": 1598.1757633019997,
  "logging_calls": 100000,
  "counter_calls": 100001,
  "logging_avg_ms": 16.58567538348,
  "counter_avg_ms": 15.981597817041027
}
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$
```

restart again:

```
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ docker compose restart
[+] Restarting 3/3
✓ Container facade-service Started
✓ Container logging-service Started
✓ Container counter-service Started
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ sleep 4
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ curl -s http://localhost:8000/accounts ; echo # should print {"balances":{}}
{"balances":{}}
```

10 clients directed at a single target account:

```

balances {}
(.venv_APZ_general) julfy@julfy-Nitro-AN515-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$ python3 perf_test.py --scenario B --clients 10 --txns 10000

=== Scenario B: shared account ===
clients=10, txns/client=10000, total=100000
client 9 (runB-shared): 2500/10000
client 0 (runB-shared): 2500/10000
client 2 (runB-shared): 2500/10000
client 4 (runB-shared): 2500/10000
client 8 (runB-shared): 2500/10000
client 6 (runB-shared): 2500/10000
client 3 (runB-shared): 2500/10000
client 7 (runB-shared): 2500/10000
client 1 (runB-shared): 2500/10000
client 5 (runB-shared): 2500/10000
client 9 (runB-shared): 5000/10000
client 0 (runB-shared): 5000/10000
client 2 (runB-shared): 5000/10000
client 8 (runB-shared): 5000/10000
client 4 (runB-shared): 5000/10000
client 6 (runB-shared): 5000/10000
client 3 (runB-shared): 5000/10000
client 7 (runB-shared): 5000/10000
client 5 (runB-shared): 5000/10000
client 1 (runB-shared): 5000/10000
client 9 (runB-shared): 7500/10000
client 0 (runB-shared): 7500/10000
client 2 (runB-shared): 7500/10000
client 8 (runB-shared): 7500/10000
client 4 (runB-shared): 7500/10000
client 3 (runB-shared): 7500/10000
client 7 (runB-shared): 7500/10000
client 6 (runB-shared): 7500/10000
client 5 (runB-shared): 7500/10000
client 1 (runB-shared): 7500/10000
client 9 (runB-shared): 10000/10000
client 0 (runB-shared): 10000/10000
client 8 (runB-shared): 10000/10000
client 2 (runB-shared): 10000/10000
client 4 (runB-shared): 10000/10000
client 7 (runB-shared): 10000/10000
client 6 (runB-shared): 10000/10000
client 3 (runB-shared): 10000/10000
client 5 (runB-shared): 10000/10000
client 1 (runB-shared): 10000/10000
wall_time      = 282.94 s
successful txns = 100000
failed txns    = 0
throughput     = 353.4 req/s
logging cumulative = 1708.23 s (avg 17.08 ms over 100000 calls)
counter cumulative = 1658.52 s (avg 16.59 ms over 100000 calls)
final balances  = {'runB-shared': 100000.0}
(.venv_APZ_general) julfy@julfy-Nitro-AN515-S7:~/Documents/6th_term/APZ/Task_1-Microservices_Basics_my_test/APZ_labs/micro_basics$

```

Metric	Scenario A (distinct accounts)	Scenario B (shared account)
Total transactions	100 000	100 000
Failed	0	0
Wall time	272.72 s	282.94 s
Throughput	366.7 req/s	353.4 req/s
Logging cumulative	1658.53 s	1708.23 s
Logging avg / call	16.59 ms	17.08 ms
Counter cumulative	1598.17 s	1658.52 s
Counter avg / call	15.98 ms	16.59 ms
Final balances	all 10 000	shared = 100 000

code: https://github.com/Bohdanok/APZ_labs/tree/micro_basics